screen, or in the water intake.

Check the raw-water filter. If you find a lot of silt, the heat exchanger (or engine itself on raw-water cooled boats) may be silted up. Feel the freshwater inlet and outlet pipes to the heat exchanger—if it is doing its job, there should be a noticeable fall in the temperature of the water leaving the heat exchanger. Many heat exchangers have removable end caps and can be rodded out with a suitable wooden dowel. If a refrigeration condenser is fitted in-line with the engine cooling circuit, any obstruction in the condenser will reduce the raw water flow to the engine.

Check for collapsed cooling hoses or poor pump performance. This is another reason that a high-set overboard discharge is useful, as the pump flow can be gauged at a glance. In certain instances, if the raw-water intake is not set low enough in the hull, a well-heeled sailboat can suck in air and air-lock the raw-water pump.

Where the raw water is injected into the exhaust (on both raw-water and heat-exchanger cooled engines) a relatively small nozzle is sometimes used to direct the water down the exhaust pipe and away from the exhaust manifold. If scale forms in the raw-water circuit, this nozzle is likely to plug, restricting the water flow.

If a boat has been operating in salt water then moves into fresh water, scale formed in salt water will swell and the engine will gradually overheat. On heat-exchanger engines with galvanized exhaust piping, check the pipes for partial blockages. On raw-water engines, first look at the water side of the exhaust manifold—a major descaling may be in order.

If the raw-water flow is normal, check any header tank as above, but *only after allowing the engine to cool*.

Perhaps the engine is overloaded (a rope around the propeller, an oversized propeller, a badly fouled bottom, too much auxiliary equipment). Maybe the ambient water temperature is higher than normal. A boat moving into the tropics may experience a 40 °F (22 °C) rise in water temperature, and engine temperatures may rise a little.

The thermostat may be malfunctioning (some raw-water engines don't have thermostats). It will be found under a bell-housing near the top and front of the engine. Take it out and try operating without it. This will make most engines run cool, but will cause

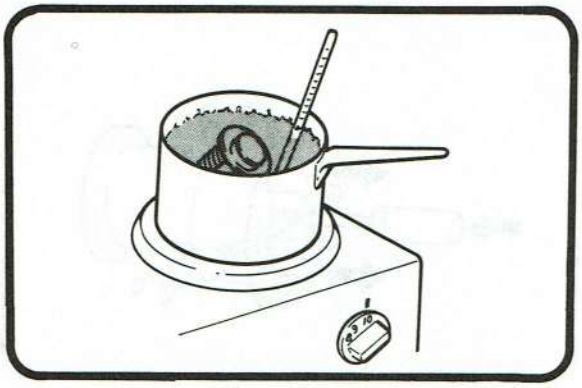


Figure 5-3. *Checking the operation of the thermostat. (Courtesy Volvo Penta)*

a few engines—e.g., some Caterpillars—to run hot and in this case should not be done. To test the thermostat, put it into a pan of water and heat it (see Figure 5-3). It should open between 165 °F and 185 °F (74 °C to 82 °C), except on some Caterpillars, which open as high as 192 °F (89 °C), and some raw-water engines, which open between 140 °F and 160 °F (60 °C and 71 °C).

Raw-water-cooled engines are likely to develop scale around the cylinders over time, especially if run above 160 °F (71 °C). The best defense against scale formation is to have adequate sacrificial zincs in the raw-water side of the cooling system, and *to change them as soon as they are partially eaten away*. This is one of the most important maintenance items on marine diesels. Descaling, when necessary, requires professional advice (see also page 93).

Faulty injection and injection dribble can cause late burning of the fuel, which will heat up the cylinders at the end of the power stroke and would normally be accompanied by black smoke. This heat is not converted into work and must be removed by the cooling system, leading to higher temperatures. Scale, or partial seizures (see below), can generate hot spots that cause pockets of steam to build up in the cylinder or head. These can sometimes air-lock cooling passages, the cooling pump, the heat exchanger, or the expansion tank, especially if the piping runs have high spots where steam or air can gather. (Note that where a domestic water heater is plumbed into the engine cooling circuit, as most are, there is frequently a high